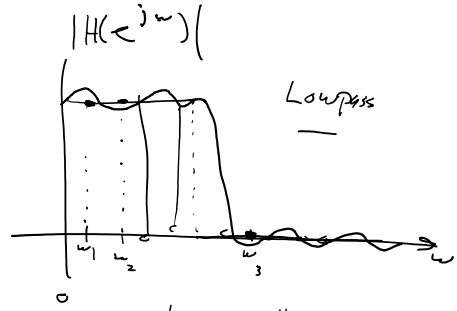
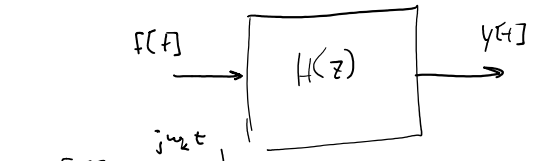
Freq Domain Filter Design

$$\boxed{\begin{aligned} y[t] + a_1 y[t-1] + \cdots + a_p y[t-p] \\ = b_0 f[t] + b_1 f[t-1] + \cdots + b_q f[t-q] \end{aligned}}$$

$$A(z) = 1 + \sum_{i=1}^{p} a_i z^{-1} \qquad B(z) = \sum_{i=0}^{q} b_i z^{-1}$$

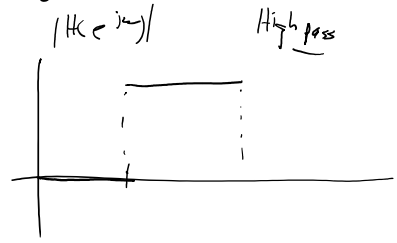
$$H(z) = \frac{B(z)}{A(z)}$$





$$\boxed{f[t] = e^{j\omega_k t}}$$

$$y[t] = \underbrace{e^{j\omega_k t}}\ \underbrace{H(e^{j\omega_k})}_{g_k}$$



$$\boxed{y[t] = g_k e^{j\omega_k t}}$$

$$g_k e^{j\omega_k t} + g_k e^{j\omega_k (t-1)} a_1 + \cdots + g_k e^{j\omega_k (t-p)} a_p$$
$$= b_0 e^{j\omega_k t} + b_1 e^{j\omega_k (t-1)} + \cdots + b_q e^{j\omega_k (t-q)}$$

$$\downarrow$$

$$g_k + a_1 e^{-j\omega_k} + a_2 g_k e^{-j2\omega_k} + \cdots + a_p g_k e^{-jp\omega_k}$$
$$= b_0 + b_1 e^{-j\omega_k} + b_2 e^{-j2\omega_k} + \cdots + b_q e^{-jq\omega_k}$$

$$\begin{array}{l} k=1 \\ k=2 \\ \\ \\ \end{array}
\begin{bmatrix} g_1 e^{-j\omega_1} & g_1 e^{-j2\omega_1} & \cdots & g_1 e^{-jp\omega_1} & 1 & e^{-j\omega_1} & \cdots & e^{-jq\omega_1} \\ g_2 e^{-j\omega_2} & g_2 e^{-j2\omega_2} & \cdots & g_2 e^{-jp\omega_2} & 1 & e^{-j\omega_2} & \cdots & e^{-jq\omega_2} \\ \vdots & & & & & & & \\ & & & & & & & \end{bmatrix}
\begin{bmatrix} -a_1 \\ -a_2 \\ \vdots \\ -a_p \\ b_0 \\ b_1 \\ \vdots \end{bmatrix}
=
\begin{bmatrix} g_1 \\ g_2 \\ \\ \\ \end{bmatrix}
+
\begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ \end{bmatrix}$$

$$k = N \quad \underbrace{\begin{bmatrix} \vdots & & & \\ \cdots & \cdots & \cdots & \cdots \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_q \end{bmatrix}}_{\underline{c}} = \underbrace{\begin{bmatrix} \\ \\ q_N \end{bmatrix}}_{x} + \underbrace{\begin{bmatrix} \vdots \\ e_N \end{bmatrix}}_{e}$$

$$\min \| Ac - x \|_2^2$$

$$\boxed{\underline{c} = (A^H A)^{-1} A^H \underline{x}}$$

Linear Predictors

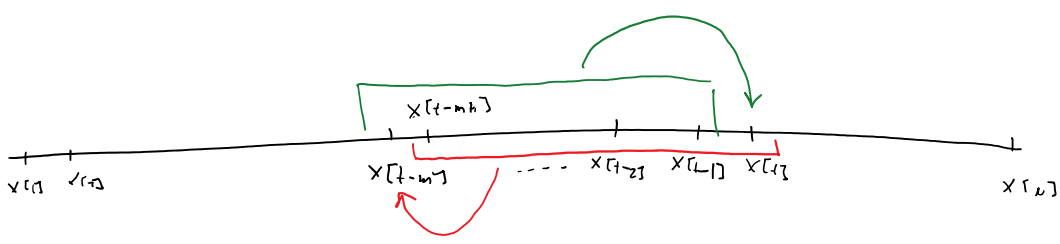
$$x[t] = -\sum_{i=1}^{m} a_i \, x[t-i] + f[t]$$

$$= -\left(a_1 x[t-1] + a_2 x[t-2] - \cdots + a_m x[t-m] \right) + f[t]$$

$$\begin{bmatrix} x[m+1] \\ x[m+2] \\ \vdots \\ x[N] \end{bmatrix} = \begin{bmatrix} x[m] & x[m-1] & \cdots & x[1] \\ x[m+1] & x[m] & \cdots & x[2] \\ & & & \\ x[N-1] & x[N-2] & \cdots & x[N-m+1] \end{bmatrix} \begin{bmatrix} -a_1 \\ -a_2 \\ \vdots \\ -a_m \end{bmatrix} + \begin{bmatrix} f[m+1] \\ f[m+2] \\ \\ f[N] \end{bmatrix}$$

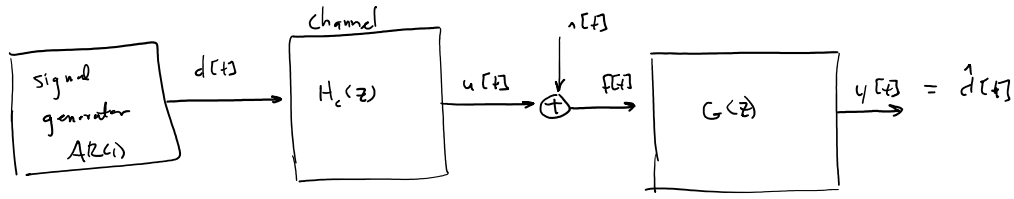
$$\underline{x} = A \underline{c} + \underline{e}$$

$$\underline{c} = (A^H A)^{-1} A^H \underline{x}$$



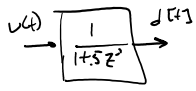
$$x[t+m] = -\left(\sum_{i=1}^{m} b_i\, x[t-m+i]\right) + b[t]$$

$$= -\left(b_1\, x[t-m+1] + b_2\, x[t-m+2] + \dots b_m\, x[t]\right) + b[t]$$

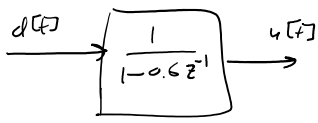
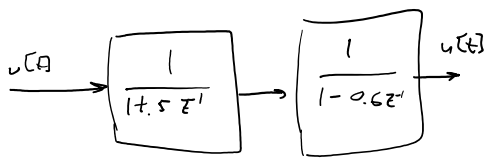


Assume: $d[t]$ is an AR(1) signal.

$$d[t] = -.5\, d[t-1] + v[t]$$

- 

- $$H_c(z) = \frac{1}{1-0.6z^{-1}}$$



$$u[t] = 0.6\, u[t-1] + d[t]$$